DineVina — Vision Document (PA2)

Performed by: Võ Mỹ Ngọc (Vision source), Nguyễn Trung Kiên (product overview source), Hoàng Cao Phong (integration) | **Reviewed by:** Lương Thiện Nhân and Võ Mỹ Ngọc (assigned; review records not supplied) | **Edited by:** Hoàng Cao Phong

Group 08 · initial Vision · 29 October 2026

1. Introduction

Performed by: Võ Mỹ Ngọc | **Reviewed by:** Lương Thiện Nhân (assigned; review record not supplied) | **Edited by:** Võ Mỹ Ngọc

This Vision Document defines the high-level needs, market position, and product capabilities of DineVina, a responsive web marketplace for restaurant discovery, table reservation requests, and event inquiries. It gives the project team and stakeholders a shared understanding of what the product should achieve, why its capabilities are needed, and which users benefit from them. Detailed functional behavior, acceptance criteria, and implementation decisions will be refined in the use-case model, supplementary specifications, and product backlog.

DineVina follows a B2B2C model in which restaurant partners publish and manage dining services, customers find suitable venues and submit requests, and the platform connects both sides through visible request states. An ordinary table reservation remains a request until the restaurant partner confirms it, while event inquiries follow a separate clarification and proposal process. The current product scope excludes a complete restaurant point-of-sale, inventory, accounting, and payment system, and it does not include native mobile applications.

1.1 References

Performed by: Võ Mỹ Ngọc | **Reviewed by:** Lương Thiện Nhân (assigned; review record not supplied) | **Edited by:** Võ Mỹ Ngọc

Reference	Date	Publishing organization or source	Use in this document
PA1 Report - DineVina: Smart Restaurant Reservation and Event Booking System	17 October 2026	Group 08, Introduction to Software Engineering	Defines the agreed product idea, boundaries, actors, nine functional groups, AI direction, and PasGo/Eatigo survey findings.
Project Assignment 2 (PA2-2026)	15 October 2026	CS300/CSC13002, University of Science, VNU-HCM	Defines the required Vision Document sections, paragraph depth for product features, Mermaid workflow requirement, and submission format.
Vision Document for a Small Project (`rup_vision_sp.docx`)	Course-provided template	Rational Unified Process template distributed with PA2	Provides the required section structure and the formats for the Problem Statement and Product Position Statement.
Vision Document Example (`rup_vision_sp-example.docx`)	Course-provided example	Distributed with PA2	Illustrates how the positioning tables and product feature section may be completed.

2. Positioning

Performed by: Võ Mỹ Ngọc | **Reviewed by:** Lương Thiện Nhân (assigned; review record not supplied) | **Edited by:** Võ Mỹ Ngọc

2.1 Problem Statement

Performed by: Võ Mỹ Ngọc | **Reviewed by:** Lương Thiện Nhân (assigned; review record not supplied) | **Edited by:** Võ Mỹ Ngọc

	Statement
The problem of	finding a restaurant that fits a customer's location, cuisine, budget, party size, capacity, and event needs while keeping restaurant information and request status clear and traceable
affects	customers planning ordinary meals or group events, restaurant partners receiving and managing requests, and platform administrators responsible for trustworthy listings
the impact of which is	customers spend time comparing fragmented or uncertain information, may confuse a submitted request with a confirmed booking, and must coordinate complex events through unstructured channels; restaurant partners face incomplete requests, conflicting capacity commitments, and inefficient follow-up
a successful solution would be	a responsive marketplace that presents attributable venue and menu information, supports relevant search criteria, separates requested and confirmed states, provides a structured event proposal path, and gives restaurant partners a single workspace for availability and incoming demand

2.2 Product Position Statement

Performed by: Võ Mỹ Ngọc | **Reviewed by:** Lương Thiện Nhân (assigned; review record not supplied) | **Edited by:** Võ Mỹ Ngọc

	Statement
For	diners looking for suitable restaurants, customers planning group events, restaurant partners seeking qualified demand, and platform administrators maintaining listing quality
Who	need a clearer way to discover venues, evaluate dining or event suitability, submit complete requests, and track responses without confusing a request with a confirmed commitment
The product name	DineVina is a responsive B2B2C web marketplace for restaurant discovery, table reservation requests, and event inquiries
That	connects customers and restaurant partners through approved venue information, menus, availability, explicit request states, structured event briefs and proposals, promotions, and preference-based assistance
Unlike	relying on separate search, social, messaging, and booking tools, or the limited PasGo and Eatigo interface flows observed in the PA1 survey
Our product	combines a traceable booking state, a dedicated event negotiation workflow, and explainable recommendations grounded in partner-verified venue data, while keeping ordinary filters available as a fallback

3. Stakeholder and user descriptions

Performed by: Hoàng Cao Phong (integration from PA1 and PA2 working draft) | **Reviewed by:** Võ Mỹ Ngọc (assigned; review record not supplied) | **Edited by:** Hoàng Cao Phong

3.1 Stakeholder and user summary

Performed by: Hoàng Cao Phong | **Reviewed by:** Võ Mỹ Ngọc (assigned; review record not supplied) | **Edited by:** Hoàng Cao Phong

Group	Interest and main use
Customers	Find a suitable approved venue, compare menus and conditions, request a table or event, and track a response.
Restaurant partners	Maintain venue and menu information, manage availability, answer requests, and prepare event proposals.
Platform administrators	Approve listings, moderate published content, and investigate platform issues.
Project team and course reviewers	Trace requirements to tasks, builds, tests, and a demonstrable increment.

Customers use mobile or desktop browsers, often with intermittent mobile data. Partner staff primarily use a desktop dashboard and may check request status on a phone. Administrators use an authenticated browser. Browsing should work without device-location permission; a manual location filter remains available. Contact and identity rules for submitting a request are implementation decisions that must be specified before release.

3.2 Key needs and priority

Performed by: Hoàng Cao Phong | **Reviewed by:** Võ Mỹ Ngọc (assigned; review record not supplied) | **Edited by:** Hoàng Cao Phong

Need	Priority	Product response
Narrow venues by location, cuisine, budget, party size and amenities	High	Search, filters and approved venue details.
Understand what a venue offers	High	Menu, capacity context, conditions and update dates.
Distinguish a request from a booking	High	Requested, confirmed, declined and cancelled states with history.
Coordinate a group event	Medium	Structured brief, clarification and proposal workflow.
Avoid conflicting partner commitments	High	Availability controls and a response queue.
Receive explainable suggestions	Medium	Attribute-based ranking and manual-filter fallback.

3.3 Alternatives and competition

Performed by: Lương Thiện Nhân (PA1 survey) and Hoàng Cao Phong (PA2 integration) | **Reviewed by:** Võ Mỹ Ngọc (assigned; review record not supplied) | **Edited by:** Hoàng Cao Phong

The PA1 survey documented PasGo's venue discovery, detail, booking form and consultation entry, and Eatigo's time-based search, discount presentation and offer conditions. Those captures establish visible interface patterns; they do not verify a completed reservation or internal processing rules. DineVina's proposed difference is an explicit request-to-confirmation state and a dedicated event inquiry path supported by approved venue data. The PA1 survey remains the evidence source for this comparison.

4. Product overview

Performed by: Nguyễn Trung Kiên (source Product Overview) and Hoàng Cao Phong (integration) | **Reviewed by:** Võ Mỹ Ngọc (assigned; review record not supplied) | **Edited by:** Hoàng Cao Phong

DineVina is a B2B2C responsive web marketplace with customer, restaurant-partner and platform-administrator roles. Its planned client, API and data tiers separate browser presentation, business rules and persistent records. The current repository contains a React/TypeScript frontend and FastAPI backend scaffold; the complete venue, reservation and event workflows remain implementation targets. A health endpoint or contract document alone does not demonstrate those workflows.

The intended authoritative records include approved venue profiles, menus, availability slots, reservation requests, event conversations and state history. The platform does not require a restaurant POS, inventory or accounting system, a payment gateway, or native mobile applications for PA2. External map and notification services are possible later dependencies, not prerequisites for the initial documentation. Manual location entry and in-app status display provide fallbacks.

The most important dependencies are accurate partner-supplied venue data, administrator approval, agreed reservation/cancellation policy and role-based access. Demo records must be approved partner data or explicitly identified as samples. A recommendation should rank only eligible venues and explain the attributes behind a result. The system boundaries and architectural options were drawn from Kiên's `research/PA2_Kien_ProductOverview` contribution; this integrated overview avoids treating proposed technology as deployed behavior.

5. Product Features

Performed by: Võ Mỹ Ngọc | **Reviewed by:** Lương Thiện Nhân (assigned; review record not supplied) | **Edited by:** Võ Mỹ Ngọc

The following features refine the nine functional groups established in the PA1 project proposal. They describe externally visible product capabilities and user value without committing the team to a particular implementation design.

5.1 Accounts and Profiles

Performed by: Võ Mỹ Ngọc | **Reviewed by:** Lương Thiện Nhân (assigned; review record not supplied) | **Edited by:** Võ Mỹ Ngọc

DineVina allows customers and restaurant partners to create accounts and maintain the contact and preference information needed for their activities. Role-aware access separates customer, partner, and administrator actions so that each user sees only the functions and records relevant to that role. Customers benefit from reusable booking details and preferences, while partners and administrators benefit from attributable actions and controlled access to operational data.

5.2 Restaurant Onboarding

Performed by: Võ Mỹ Ngọc | **Reviewed by:** Lương Thiện Nhân (assigned; review record not supplied) | **Edited by:** Võ Mỹ Ngọc

Restaurant partners can submit a venue profile containing its identity, location, cuisine, photos, amenities, capacity, and contact details. A platform administrator reviews the submission and either approves it or returns it for correction before it becomes an attributable source for customer discovery. This feature gives customers more trustworthy venue information and helps partners present their restaurants consistently on the marketplace.

5.3 Restaurant Discovery and Venue Details

Performed by: Võ Mỹ Ngọc | **Reviewed by:** Lương Thiện Nhân (assigned; review record not supplied) | **Edited by:** Võ Mỹ Ngọc

Customers can search and filter restaurants by criteria such as location, cuisine, budget, party size, capacity, and amenities, then open a venue page for more detail. The feature shortens the path from a dining or event need to a manageable set of suitable options instead of forcing users to compare fragmented information manually. Customers benefit from more relevant choices, while approved restaurant partners gain visibility among users whose needs match their offerings.

5.4 Menu Management and Viewing

Performed by: Võ Mỹ Ngọc | **Reviewed by:** Lương Thiện Nhân (assigned; review record not supplied) | **Edited by:** Võ Mỹ Ngọc

Restaurant partners can maintain menu items with approved names, descriptions, and prices, and customers can review this information before submitting a request. Menu visibility is needed because price and dish suitability influence whether a venue fits a customer's budget, dietary preferences, and occasion. Customers make better-informed decisions, and partners receive requests from people who have already reviewed the relevant offering.

5.5 Table Reservation Requests

Performed by: Võ Mỹ Ngọc | **Reviewed by:** Lương Thiện Nhân (assigned; review record not supplied) | **Edited by:** Võ Mỹ Ngọc

Customers can choose a date, time, and party size and submit a table reservation request to a restaurant partner. The partner can confirm or reject the request, while both sides can see its status history and applicable cancellation conditions; submitting a request does not itself create a confirmed reservation. This feature gives customers a clear path from venue selection to a visible outcome and helps restaurant staff respond to demand without making ambiguous promises.

5.6 Event Inquiries and Proposals

Performed by: Võ Mỹ Ngọc | **Reviewed by:** Lương Thiện Nhân (assigned; review record not supplied) | **Edited by:** Võ Mỹ Ngọc

Customers planning an event can submit a structured brief containing the occasion, expected headcount, preferred date, budget, and special needs. Restaurant partners can request clarification, propose terms, and track the customer's response in one conversation rather than treating a complex event like an ordinary table request. Customers benefit from a clearer planning process, and partners benefit from receiving the information required to prepare realistic proposals.

5.7 Availability and Partner Dashboard

Performed by: Võ Mỹ Ngọc | **Reviewed by:** Lương Thiện Nhân (assigned; review record not supplied) | **Edited by:** Võ Mỹ Ngọc

Restaurant partners can maintain operating hours, capacity, and blocked slots while viewing incoming reservation and event requests in a role-specific dashboard. Keeping availability and demand together is needed to reduce conflicting commitments and give staff a practical queue for daily work. Partners benefit from better operational control, while customers benefit from choices and responses based on more current partner-provided information.

5.8 Promotions and Editorial Content

Performed by: Võ Mỹ Ngọc | **Reviewed by:** Lương Thiện Nhân (assigned; review record not supplied) | **Edited by:** Võ Mỹ Ngọc

Restaurant partners can submit time-bound promotions and venue content under platform rules, with eligibility conditions and expiry information visible to customers. Clear conditions are needed so that an offer can support discovery without creating uncertainty about when or to whom it applies. Customers benefit from understandable value, partners gain a controlled promotional channel, and administrators can moderate content quality without turning DineVina into a full merchant accounting or payment system.

5.9 Intelligent Assistance

Performed by: Võ Mỹ Ngọc | **Reviewed by:** Lương Thiện Nhân (assigned; review record not supplied) | **Edited by:** Võ Mỹ Ngọc

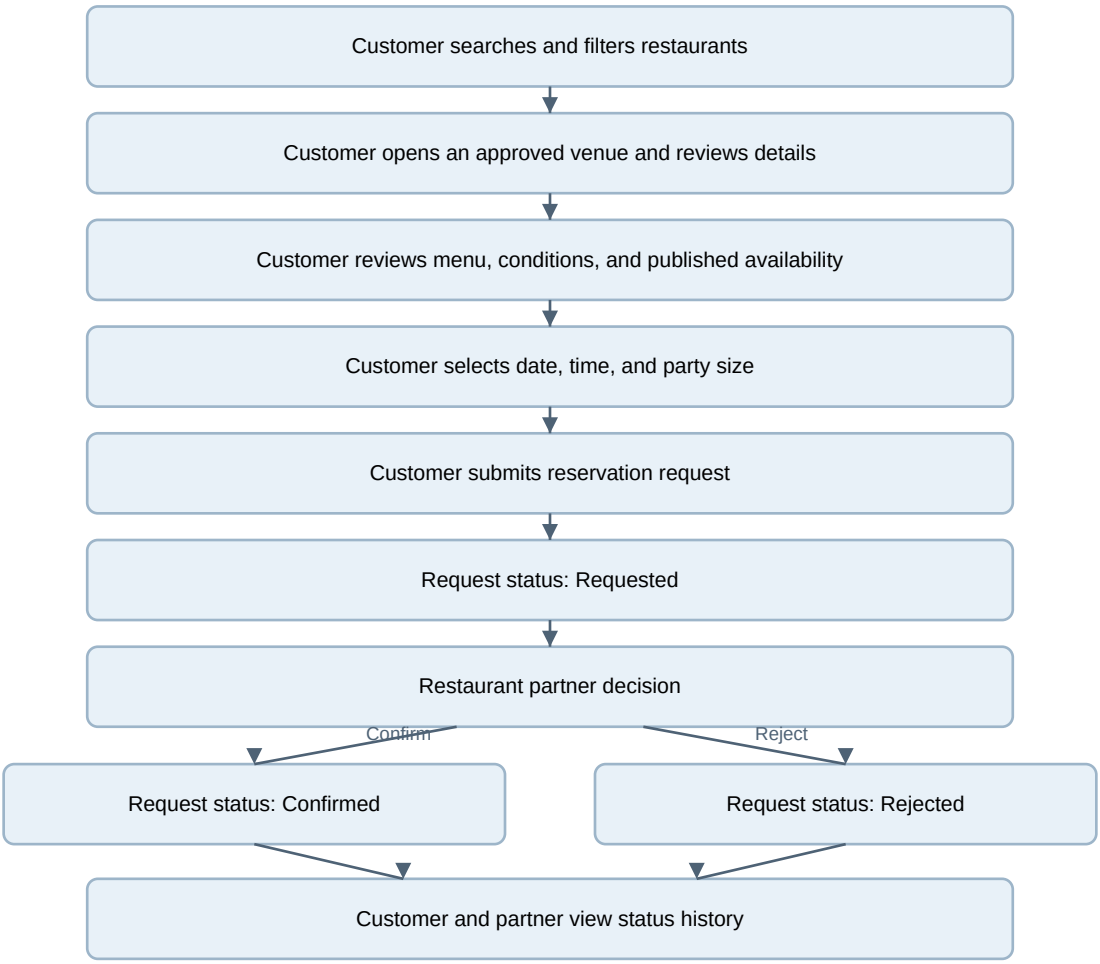
DineVina is planned to recommend restaurants by first excluding unapproved or unsuitable venues and then ranking eligible choices using customer preferences and verified venue attributes such as cuisine, location, budget, occasion, and menu information. Each suggestion should include visible reasons, while ordinary search and filters remain available when recommendations are unavailable or unhelpful. This capability is intended to reduce search effort for customers and improve qualified exposure for partners, but it remains a proposed feature rather than an implemented capability in the current project scaffold.

5.10 Key User Workflows

Performed by: Võ Mỹ Ngọc | **Reviewed by:** Lương Thiện Nhân (assigned; review record not supplied) | **Edited by:** Võ Mỹ Ngọc

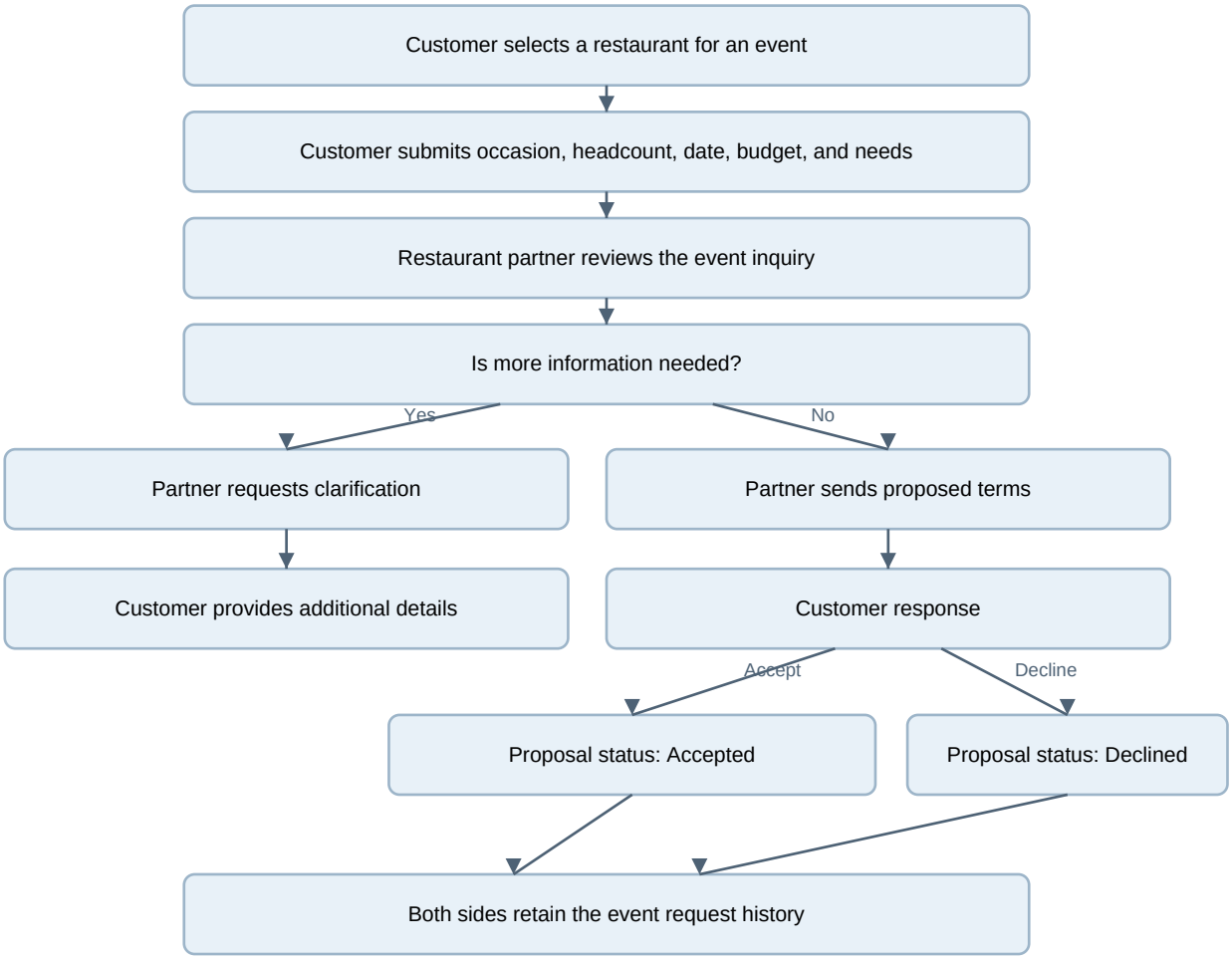
Table Reservation Request

Performed by: Võ Mỹ Ngọc | **Reviewed by:** Lương Thiện Nhân (assigned; review record not supplied) | **Edited by:** Võ Mỹ Ngọc



Event Inquiry and Proposal

Performed by: Võ Mỹ Ngọc | **Reviewed by:** Lương Thiện Nhân (assigned; review record not supplied) | **Edited by:** Võ Mỹ Ngọc



Clarification and revision return to partner review.

6. Non-functional requirements

Performed by: Hoàng Cao Phong (integration from PA2 working draft) | **Reviewed by:** Võ Mỹ Ngọc (assigned; review record not supplied) | **Edited by:** Hoàng Cao Phong

The values below are proposed acceptance targets. They require measurement on an agreed test environment before the team can claim achievement.

Quality attribute	Acceptance target or verification method
Performance	At least 95% of search and venue-detail API requests finish within 800 ms at 100 concurrent virtual users on the documented test environment; first useful page content within 2.5 s on a defined mid-range mobile/4G profile.
Security	Server-side role authorization on all partner/admin mutations; HTTPS for public deployment; password hashes stored with a maintained password-hashing library; no secrets in Git.
Privacy	Use only contact and booking data needed for service; exclude real customer records from repository and demo fixtures; agree retention and deletion rules before public use.

Quality attribute	Acceptance target or verification method
Booking reliability	A concurrent capacity test must prevent confirmed reservations above a slot limit; every state change is timestamped.
Accessibility	Core search and request flows support keyboard use, labelled inputs and visible focus; inspect against WCAG 2.2 AA.
Platform	Test current Chrome, Firefox, Edge and Safari at 360 px mobile and 1280 px desktop widths.
Maintainability	Backend tests/lint and frontend production build pass at each tested build; API contract changes receive peer review.
Recommendation quality	Evaluate at least 30 representative preference queries and report precision at 3, unsuitable-result count, coverage and failure examples.
Recovery	Demonstrate a database backup and restore on a test copy before the final demo; record elapsed restoration time.

References

Performed by: Hoàng Cao Phong | **Reviewed by:** Võ Mỹ Ngọc (assigned; review record not supplied) | **Edited by:** Hoàng Cao Phong

- Course Project Assignment 2 (PA2-2026) and the course Vision template.
- Group 08 PA1 project proposal and PasGo/Eatigo survey.
- Source contributions: Ngọc, `research/PA2_Ngoc_Vision_Overview`; Kiên, `research/PA2_Kien_ProductOverview`; Group 08 PA2 working Vision draft.
- WCAG 2.2 for the accessibility target.